

Network Layer: IPv4 Static Routing

Lecture 11 | CSE421 – Computer Networks

Department of Computer Science and Engineering
School of Data & Science

Objectives

Static Routing

- Standard static routing
  - Directly Attached / Connected
  - Next Hop / Recursive
- Default Routing
- Floating Static Routing

Learning About Networks

A router can learn about remote networks in one of two ways:

- Manually - Remote networks are manually entered into the route table using static routes.
- Dynamically - Remote networks are automatically learned using a dynamic routing protocol.

Learning About Networks

BRAC UNIVERSITY
Inspiring Excellence

Table of R1

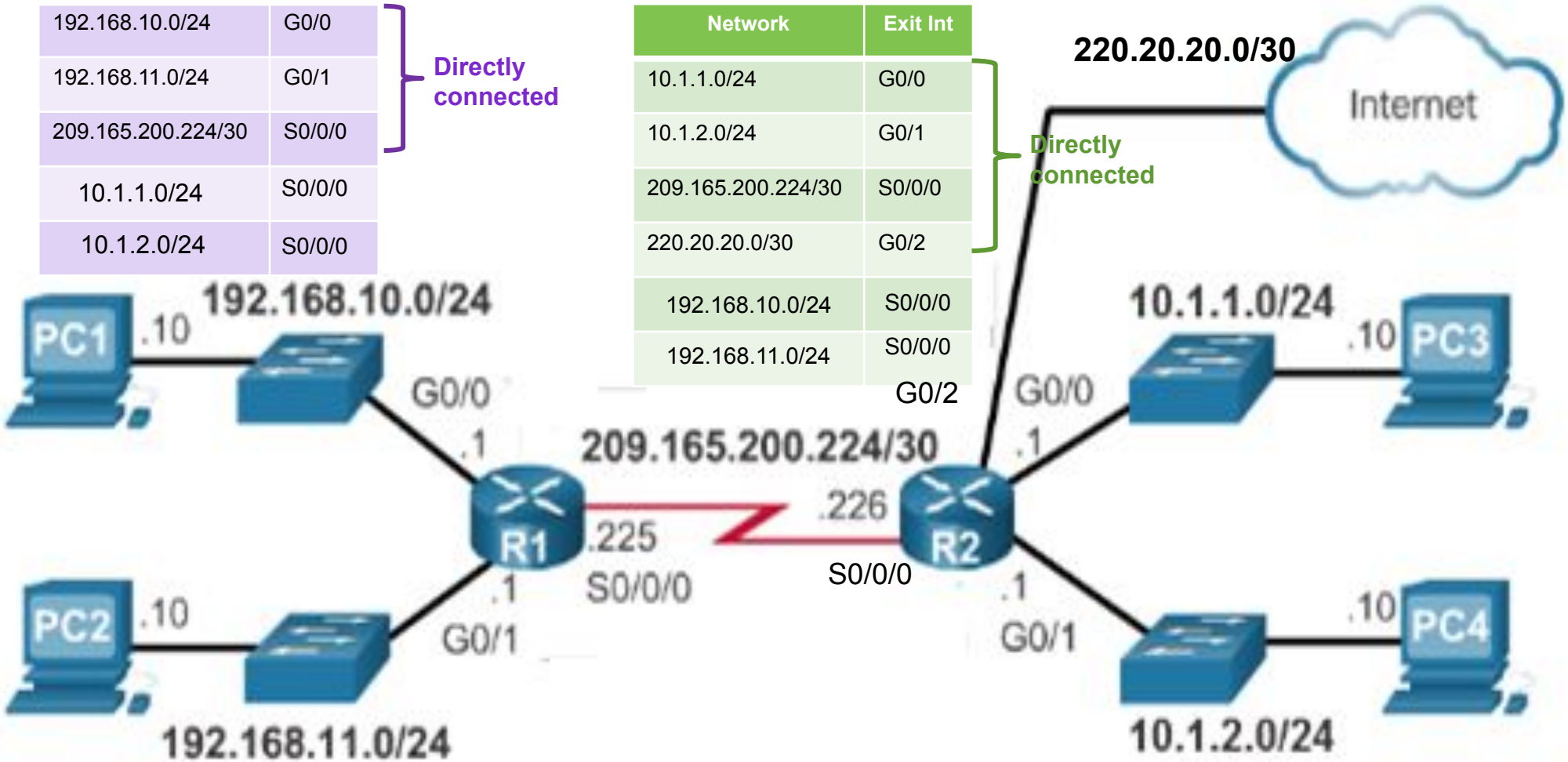
Network	Exit Int
192.168.10.0/24	G0/0
192.168.11.0/24	G0/1
209.165.200.224/30	S0/0/0
10.1.1.0/24	S0/0/0
10.1.2.0/24	S0/0/0

Directly connected

Table of R2

Network	Exit Int
10.1.1.0/24	G0/0
10.1.2.0/24	G0/1
209.165.200.224/30	S0/0/0
220.20.20.0/30	G0/2
192.168.10.0/24	S0/0/0
192.168.11.0/24	S0/0/0

Directly connected



Static Route

- Manually configured route
- Used to reach networks not directly connected
- Does not update automatically
- Changes must be applied manually

Static Route Advantages

- Simple and predictable
- More secure (not advertised)
- Low CPU/memory usage

Static Route Disadvantages

- Time-consuming to configure
- Not scalable for large networks
- No automatic failover
- Requires full network knowledge

BRAC UNIVERSITY
Inspiring Excellence

The End